Building Strategic Resilience: Lessons from an Industrial Automation Company

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Abstract

Industrial companies increasingly operate in environments shaped by geopolitical instability, supply chain disruptions, and economic uncertainty. These events have highlighted the importance of strategic resilience, defined as an organization's ability to sustain value creation despite major external shocks. This paper examines strategic resilience through the case of a French manufacturer of high-precision automated systems for the fragrance industries. Although the company entered the COVID-19 pandemic from a position of financial strength and technological leadership, the crisis exposed several structural dependencies embedded within its business model. The analysis identifies these vulnerabilities and discusses how market diversification, greater customer autonomy, and data-driven services can reduce exposure to future disruptions. The case illustrates that resilience is not solely determined by operational performance or financial health, but also by the adaptability of the business model itself.

Introduction

Operational excellence has long been a source of competitive advantage for industrial companies. Continuous improvement, automation, and globalization have enabled organizations to increase productivity, reduce costs, and expand into international markets. However, these same choices have also increased dependence on complex supply chains, global mobility, and interconnected operations.

The COVID-19 pandemic demonstrated how quickly these dependencies could become strategic vulnerabilities. Companies with strong financial performance and efficient operations still faced significant disruption when travel restrictions, supply chain interruptions, and market uncertainty altered the conditions on which their business models relied. Indeed, the crisis renewed attention to strategic resilience, defined as an organization's capacity to sustain value creation despite major external shocks.

This paper examines strategic resilience through the case of Company A, a French manufacturer of high-precision automated dispensing systems for the fragrance and flavor industries. The company entered the pandemic in a strong competitive position. Yet, the crisis exposed several structural vulnerabilities within its business model. The analysis identifies these dependencies and discusses practical actions that can strengthen resilience before the next macroeconomic shock.

Findings and Analysis

Company Overview

Founded in 1996 and headquartered in Montpellier, France, Company A designs and manufactures high-precision automated dispensing systems for the fragrance and food flavor industries. Its solutions automate the weighing, dosing, storage, and transfer of liquids, powders, and pastes, enabling customers to improve productivity, reproducibility, and product quality. In addition to its equipment, the company provides software, integration services, and long-term maintenance contracts, allowing it to build lasting relationships with its customers.

By 2018, Company A had established itself as a global leader in its niche market, with more than 180 installations worldwide and approximately 23% of its revenue generated from exports. The company operated in a favourable competitive environment, characterised by a limited number of international competitors, high barriers to entry, and relatively low pressure on margins. Its differentiated positioning relied on highly customised solutions that integrated seamlessly into customers' production lines, supported by a strong reputation for precision, reliability, and performance.

Several internal capabilities reinforced this competitive position. Company A developed a modular product architecture based on standard building blocks that could be assembled into customised systems. This approach simplified manufacturing, procurement, installation, and maintenance while supporting continuous product improvement through customer feedback. The company also maintained an excellent financial position, combining sustained organic growth, high profitability, and very limited debt. Between 2016 and 2018, revenues increased by 24% annually, operating income by 44%, and net income by 40%, reaching a net profit margin of 27%.

Impact of the COVID-19 Pandemic

In early 2020, the COVID-19 pandemic disrupted global trade and international mobility on an unprecedented scale. Governments introduced travel restrictions, closed borders, and implemented health measures that severely limited the movement of people and goods. These disruptions affected organizations across virtually every industry and forced many companies to reassess the resilience of their operating models.

For Company A, the immediate challenge was not a decline in demand. The fragrance and food flavor industries remained relatively resilient, and broader industrial trends continued to support automation as manufacturers sought to reduce their reliance on manual labour and improve operational continuity. At the same time, advances associated with the Fourth Industrial Revolution continued to accelerate the adoption of connected and automated production systems.

The company's primary constraint came from its international operations. Border closures prevented technical teams from travelling outside Europe to install new equipment and provide on-site support to customers. This raised a strategic question that extended well beyond the immediate effects of the pandemic: How can Company A continue to thrive in a context where international mobility can no longer be taken for granted?

Strategic Vulnerabilities

The pandemic did not fundamentally weaken Company A's competitive position. The company continued to operate in attractive markets, benefited from strong customer relationships, and maintained healthy financial performance. Instead, the crisis exposed several structural dependencies embedded within its business model. These dependencies had supported the company's growth for many years but became sources of vulnerability when international mobility could no longer be assumed.

The first vulnerability concerned the company's dependence on international mobility. Company A's business model relied heavily on engineers travelling to customer sites to install new equipment and resolve complex technical issues. Border closures disrupted this model almost overnight, preventing the company from supporting customers outside Europe through its traditional operating model. Although Company A quickly adapted by shipping spare parts directly to customers, the pandemic demonstrated that business continuity remained closely tied to physical travel.

The second vulnerability related to market concentration. Company A operated successfully within the fragrance and food flavour industries, both of which remained relatively resilient during the pandemic. Nevertheless, concentrating a significant share of revenues within a limited number of industries and export markets increased the company's exposure to future disruptions affecting those sectors or regions. Financial performance therefore depended not only on operational excellence but also on external conditions over which the company had little control.

Finally, the analysis revealed an untapped strategic asset. Company A generated valuable operational data through its installed base but made very limited use of this information. The business model remained centred on the sale, installation, and maintenance of physical equipment, despite the broader shift towards connected manufacturing, IoT, and data-driven services associated with the Fourth Industrial Revolution. This represented a missed opportunity to strengthen customer relationships, reduce dependence on on-site interventions, and create additional sources of value beyond the machines themselves.

Taken together, these findings suggest that Company A's main challenge was not operational efficiency but business model resilience. The pandemic acted as a stress test, revealing which assumptions remained valid and which dependencies could limit the company's ability to respond to future macroeconomic shocks.

Key Findings

The analysis highlights several important findings. First, Company A entered the COVID-19 crisis from a position of strength. The company was financially sound, operated in an attractive niche market, and had established a strong competitive position through its technological expertise and customer relationships. The challenges it encountered were therefore not the result of poor operational performance or weak financial fundamentals.

Second, the pandemic exposed vulnerabilities that originated from the company's business model rather than its products or capabilities. Dependence on international mobility, concentration in a limited number of markets, and the limited use of operational data all increased the company's exposure to external disruptions. These dependencies remained largely invisible under normal market conditions but became apparent when the external environment changed abruptly.

Finally, the case demonstrates that strategic resilience extends beyond crisis response. Organizations cannot anticipate every future disruption, but they can identify structural dependencies and reduce their exposure before the next macroeconomic shock occurs. This perspective provides the foundation for the discussion and recommendations presented in the following sections.

Discussion

Strategic resilience is built before the crisis

The Company A case illustrates an important characteristic of strategic resilience: it cannot be developed once a crisis has already begun. During periods of stability, many structural dependencies remain hidden because they do not constrain day-to-day operations. External shocks expose these dependencies by changing the assumptions on which the business model relies.

In Company A's case, the pandemic did not create new weaknesses. It revealed an existing dependence on international mobility, a concentrated market portfolio, and a business model centred on physical service delivery. These characteristics had supported the company's growth for many years, yet they also reduced its ability to adapt when the external environment changed.

This observation suggests that resilience should be considered a strategic capability rather than an operational response. Organizations that continuously assess their structural dependencies are better positioned to absorb uncertainty without compromising their long-term objectives.

Resilience requires strategic optionality

A second lesson emerging from this case is the importance of maintaining strategic options. Organizations become more resilient when they avoid relying on a single market, a single delivery model, or a single source of value creation.

Strategic optionality provides alternative paths when external conditions change. Market diversification reduces dependence on a limited number of industries or geographical regions. Remote service capabilities reduce reliance on physical interventions. Digital services create additional sources of value that complement traditional product sales. Each initiative increases the organization's flexibility and reduces the impact of future disruptions.

The objective is not to eliminate risk entirely. It is to ensure that unexpected events do not threaten the viability of the business model.

Resilience and digital transformation are closely connected

The analysis also highlights the relationship between digital transformation and strategic resilience. Digital technologies are often adopted to improve operational efficiency, increase productivity, or reduce costs. Their contribution to resilience is equally important.

Operational data, remote monitoring, predictive maintenance, and connected products enable organizations to continue supporting customers despite travel restrictions or operational disruptions. These capabilities also strengthen customer relationships and create opportunities to develop service-based business models alongside traditional product sales.

Digital transformation therefore extends beyond operational performance. It enables organizations to redesign the way they create, deliver, and capture value, making them better equipped to respond to future macroeconomic shocks.

Recommendations

The analysis suggests that Company A can strengthen its strategic resilience by reducing its dependence on a limited number of markets, increasing the autonomy of its customers, and expanding the role of digital technologies within its business model. These recommendations address the structural vulnerabilities identified in the previous chapter and aim to improve the company's ability to respond to future macroeconomic disruptions.

Diversify Markets and Revenue Sources

The first recommendation is to diversify Company A's customer base across both industries and geographical markets. Expanding into sectors with similar technical requirements would reduce dependence on the fragrance and food flavour industries while allowing the company to leverage its existing expertise and technologies. Likewise, entering new geographical markets would reduce exposure to regional disruptions and create additional growth opportunities.

Diversification should remain closely aligned with Company A's core capabilities. Targeting industries that require high-precision automated dispensing systems would minimise development costs while preserving the company's competitive advantage.

Increase Customer Autonomy

The pandemic demonstrated the limitations of a service model that depends heavily on on-site interventions. Company A should therefore continue reducing the need for physical support by enabling customers to perform a greater share of maintenance and troubleshooting activities independently.

This objective could be supported through more modular equipment, improved technical documentation, remote assistance, and guided maintenance procedures. Increasing customer autonomy would reduce operational constraints during future disruptions while improving service responsiveness and customer satisfaction.

Expand Data-Driven Services

Company A's installed base generates valuable operational data that remains largely underutilised. Developing capabilities such as remote monitoring, predictive maintenance, and performance analytics would strengthen customer support while creating new sources of value beyond equipment sales.

Over time, these capabilities could support the transition from a product-focused business model towards a more service-oriented offering. Subscription-based digital services, preventive maintenance contracts, and performance optimisation solutions would generate recurring revenue while reducing dependence on physical interventions.

Expected Benefits

Implemented together, these recommendations would improve Company A's resilience by reducing its exposure to external shocks while preserving its existing competitive strengths. Diversification would reduce dependence on individual markets, customer autonomy would decrease reliance on international mobility, and data-driven services would create a more flexible and scalable business model. Collectively, these initiatives would enable the company to respond more effectively to future disruptions without compromising long-term growth.

Conclusion

This paper examined how industrial companies can strengthen their strategic resilience through the case of Company A, a French manufacturer of automated dispensing systems. The analysis showed that the COVID-19 pandemic did not fundamentally alter the company's competitive position. Instead, it exposed structural dependencies within its business model, particularly its reliance on international mobility, concentrated markets, and physical service delivery.

These findings suggest that resilience should be viewed as a strategic capability rather than a reactive response to crises. Organizations cannot predict future macroeconomic shocks, but they can reduce their exposure by identifying critical dependencies, increasing strategic flexibility, and continuously adapting their business models to changing conditions.

For Company A, market diversification, greater customer autonomy, and the development of data-driven services represent practical opportunities to strengthen resilience while supporting long-term growth. More broadly, this case illustrates that resilience is not simply about recovering from disruption. It is about designing organizations that remain capable of creating value even when the assumptions on which they were built no longer hold.